How-To Library

Informatica

Secure@Source 4.5 Quick Start Deployment Guide: External Hadoop Cluster

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Abstract

This guide describes the installation process for Secure@Source 4.5 in a Linux environment for an external Hadoop cluster.

Supported Versions

- Secure@Source 4.5

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Steps to Deploy Secure@Source

You can deploy Secure@Source on a Hadoop cluster that you have set up on Cloudera, Hortonworks, or Microsoft Azure HDInsight. If you have enabled Kerberos authentication in your enterprise to authenticate users and services on a network, you can configure the Informatica domain to use Kerberos network authentication.

After you procure and setup the hardware according to the *Secure@Source Sizing Guide*, verify the prerequisites for installing Secure@Source. Then start the installation process and configure connectivity to the data sources that you plan to scan.

To deploy Secure@Source, complete the steps in this article.

Note: Some steps refer to sections in the *Informatica Secure@Source Installation and Configuration Guide*. You can access the guide on the Informatica Network at https://search.informatica.com/.

Step 1. Verify the Cloudera, Hortonworks, or Microsoft Azure HDInsight Version

Verify that you have a Secure@Source-supported version of Cloudera, Hortonworks, or Microsoft Azure HDInsight. To determine the supported version, see the *Secure@Source 4.5 Product Availability Matrix* located at https://network.informatica.com/docs/DOC-17804.

Step 2. Verify Prerequisites for the Secure@Source Server and Informatica Services Node

Operating System

Log in as the root user and verify the following Operating System requirements:

- ❑ A Secure@Source-supported Linux distribution is installed. To determine the supported version, see the *Secure@Source 4.5 Product Availability Matrix* located at https://network.informatica.com/docs/DOC-17804.
- ❑ The operating system license is registered.
- ❑ The following packages are installed:

❑curl ❑JAVA **Note:** Informatica recommends version jdk1.8.0_162 ❑rpm ❑scp ❑ssh	❑tar ❑unzip ❑wget ❑zip

Disk Space and Processors

Verify the following Secure@Source server prerequisites:

- ❑ Disk space for the mount point where you plan to install Secure@Source is at least 500 GB.
- ❑ The appropriate number of processors are available, based on the environment sizing agreement.
- ❑ The appropriate number of cores are available, based on the environment sizing agreement.
- ❑ The appropriate amount of memory is available, based on the environment sizing agreement.
- ❑ The appropriate disk space is available, based on the environment sizing agreement.
- ❑ The `/tmp` folder is at least 16 GB.

Create User

Create the Secure@Source user to install and run Secure@Source on the Secure@Source server.

Settings

Verify or configure the following settings:

- ❑ The `/etc/hosts` file includes the fully qualified host names of the Secure@Source server, Hadoop server, and all Hadoop nodes.
  Example: `<machine ipaddress> <fully qualified name> <alias>`
- ❑ The LANG environment variable is set to `en_US.UTF-8`.
- ❑ The LC_ALL environment variable is not set.
- ❑ The JRE_HOME environment variable is not set.
- ❑ The following user limits are set:
  - Max user processes: 100000

- Stack size: 25000
- Open files: 100000

❑ SELinux is set to Disabled on the Secure@Source node.

❑ Configure the firewall and verify the necessary ports are open. See "Step 6. Secure@Source and Hadoop Ports" on page 9.

❑ Start the ntpd service on the Secure@Source installation node.

❑ Verify that the ntpstat utility reports a synchronized status.

❑ Disable chronyd, the ntp replacement in RHEL7.x.

❑ Set all nodes to the same time zone.

❑ Ensure that you do not create the Informatica domain on a node in the existing Hadoop cluster.

❑ If you plan to install in Kerberos mode, you must install the krb5-workstation package before you enable Kerberos.

Set Environmental Variables

For Informatica Services	For Oracle	For IBM DB2	For Microsoft SQL Server
❑ JAVA_HOME ❑ LD_LIBRARY_PATH ❑ ODBCHOME ❑ ODBCINI ❑ ORACLE_HOME ❑ PATH	❑ ORACLE_HOME ❑ TNS_ADMIN= $ORACLE_HOME/network/admin/tnsnames.ora ❑ PATH must contain $ORACLE_HOME/bin ❑ LD_LIBRARY_PATH must contain $ORACLE_HOME/lib	❑ DB2_HOME ❑ PATH must contain $DB2_HOME/bin ❑ LD_LIBRARY_PATH must contain $DB2_HOME/lib	❑ ODBCHOME ❑ PATH must contain $ODBCHOME/bin ❑ LD_LIBRARY_PATH must contain $ODBCHOME/lib ❑ ODBCINST must be set to $ODBCHOME/odbcinst.ini ❑ ODBCINI must be set to $ODBCHOME/odbc.ini

Install and Configure the Database Client for the Repository Database

Oracle Settings

If the Secure@Source repository database is Oracle, complete the following steps on the Secure@Source server:

❑ Install and configure the Oracle Administrator Client software on the Secure@Source node.

❑ Verify that the `tnsnames.ora` file is readable by other users.

❑ Verify that the `tnsnames.ora` file has an entry for the repository database.

IBM DB2 Settings

If the Secure@Source repository database is IBM DB2, complete the following steps on the Secure@Source server:

❑ Install and configure the IBM DB2 Client software on the Secure@Source node.

❑ Verify that the IBM DB2 catalog entry is added for the repository database.

❑ Verify that the *NCHAR_MAPPING* variable is set to `GRAPHIC_CU16`.

Step 3. Verify the Services for the External Hadoop Cluster

You must configure Zookeeper, HDFS, and Yarn specifications when you install Secure@Source on an external Hadoop cluster in your enterprise. Catalog Service uses these specifications and launches the following services and components on the Hadoop cluster as Yarn application:

- Solr 6.5.0
- HBase 1.1.2
- Scanner components

Step 4. Verify the Prerequisites for the Existing Hadoop Cluster

Before you install Secure@Source on a Cloudera, Hortonworks, or Microsoft Azure HDInsight environment, you must verify that the system environment meets the prerequisites required to deploy Secure@Source.

Verify that the external Hadoop distribution meets the following prerequisites:

- ❑ Verify that OpenSSL is installed on the cluster nodes.
- ❑ Verify that JDK 1.8 is installed on all cluster nodes.
- ❑ Verify that the `/usr/bin/java` link points to the JDK 1.8 location.
- ❑ Verify that the secure path in the `/etc/sudoers` file has the `/usr/bin` directory location at the start.
- ❑ On each host machine, verify that you have the zip and unzip utilities available.
- ❑ Verify that the maximum number of open file descriptors is 100,000 or more. use the ulimit command to verify the current value and change the value if required.
- ❑ If you plan to stream user activity, copy UBA rpm from `/<Installer location>/Server/SATS/source/SecureAtSourceService/spark-components/` and install UBA rpm on all nodes of the cluster.
  Where `<Installer location>` is the folder containing the extracted installer files.

Hadoop Cluster is Kerberos-Enabled

Complete the following steps if the Hadoop cluster is enabled for Kerberos:

- ❑ Create the `<service cluster name>` directory under `/user` and assign the ownership of the directory to the service cluster name.
  Example:

  ```
  hdfs dfs -mkdir /user/<service cluster name>
  hdfs dfs -chown -R <service cluster name> /user/<service cluster name>
  ```

  Where the `<service cluster name>` is the same as the Kerberos principal user name.

  Ensure that the Kerberos principal user is created on all the Hadoop nodes.
- ❑ Create the `service-logs` directory under `/Informatica/LDM/<service cluster name>/` and assign the ownership of the `/Informatica` directory recursively to the service cluster name.
- ❑ Verify that you have the Read, Write, and Execute permissions for owners, groups, and others on the HDFS directories.
- ❑ Change the group of `/Informatica/LDM/<ServiceClusterName>/service-logs` to `hadoop`.
  Example:

  ```
  hdfs dfs -mkdir /Informatica
  hdfs dfs -mkdir /Informatica/LDM
  hdfs dfs -mkdir /Informatica/LDM/<Service Cluster Name>
  ```

```
hdfs dfs -mkdir /Informatica/LDM/<Service Cluster Name>/service-logs
hdfs dfs -chown -R <user_name> /Informatica
hdfs dfs -chgrp hadoop /Informatica/LDM/<Service Cluster Name>/service-logs
```

Hadoop Cluster is Not Kerberos-Enabled

Complete the following steps if the Hadoop cluster is not enabled for Kerberos:

- ❑ Create the `<Informatica domain user name>` directory under `/user` and assign the ownership of directory to the domain user.
  Example:

  ```
  hdfs dfs -mkdir /user/<Informatica domain user name>
  hdfs dfs -chown -R <Informatica domain user name> /user/<Informatica domain user name>
  ```

  Ensure that the Informatica domain user is created on all the Hadoop nodes.

- ❑ Create the `service-logs` directory under `/Informatica/LDM/<Informatica domain user name>/` and assign the ownership of the `/Informatica` directory recursively to the domain user.

- ❑ Verify that you have the Read, Write, and Execute permissions for owners, groups, and others on the HDFS directories.

- ❑ Change the group of `/Informatica/LDM/<ServiceClusterName>/service-logs` to `hadoop`.
  Example:

  ```
  hdfs dfs -mkdir /Informatica
  hdfs dfs -mkdir /Informatica/LDM
  hdfs dfs -mkdir /Informatica/LDM/<Service Cluster Name>
  hdfs dfs -mkdir /Informatica/LDM/<Service Cluster Name>/service-logs
  hdfs dfs -chown -R <user_name> /Informatica
  hdfs dfs -chgrp hadoop /Informatica/LDM/<Service Cluster Name>/service-logs
  ```

Step 5. Set up Kerberos and SSL for an External Cluster

You can install Secure@Source on an external cluster that uses Kerberos network authentication to authenticate users and services on a network. Secure@Source also supports SSL authentication for secure communication in the cluster.

Kerberos is a network authentication protocol which uses tickets to authenticate access to services and nodes in a network. Kerberos uses a Key Distribution Center (KDC) to validate the identities of users and services and to grant tickets to authenticated user and service accounts. In the Kerberos protocol, users and services are known as principals. The KDC has a database of principals and their associated secret keys that are used as proof of identity. Kerberos can use an LDAP directory service as a principal database.

Informatica does not support cross or multi-realm Kerberos authentication. The server host, client machines, and Kerberos authentication server must be in the same realm.

The Informatica domain requires keytab files to authenticate nodes and services in the domain without transmitting passwords over the network. The keytab files contain the service principal names (SPN) and associated encrypted keys. Create the keytab files before you create nodes and services in the Informatica domain.

Configure the SSL Protocol between the Domain and the External Cluster

Note: Informatica only supports the custom SSL option for an external Hadoop cluster configuration that is SSL enabled.

If the external cluster is SSL enabled, you must install Secure@Source using the custom SSL option. Perform the following steps to use the custom SSL option:

1. Generate the custom keystore and truststore certificates. For more information, see the H2L article *How to Create Keystore and Truststore Files for Secure Communication in the Informatica Domain* located on the Informatica Network: https://search.informatica.com.

2. Export the certificate from the Domain host keystore and import it to the Cloudera or Hortonworks truststore.
3. Export the certificate from the Cloudera or Hortonworks keystore and import it to the Doman host truststore.
4. If you plan to use a separate keystore file for the Apache Solr application, import the following certificates to allow the Apache Spark and the scanner jobs to connect to the Apache Solr application:
   - Import the certificates for the Apache Solr keystore to the cluster nodes truststore file configured for the Cloudera or Hortonworks truststore.
   - Import the certificate for the Apache Solr application into the truststore file of the Informatica domain.

   Note: Ignore this step if you plan to use the same keystore file for the Apache Solr application as the one used for HDFS and YARN.

   Use the following commands to export and import the certificate:

```
 keytool -export -keystore
<ac_keystore_file> -storepass <ac_keystore_password>
-file <file>.cert -alias <alias_of_ac_cert>
```

```
 keytool -import -keystore
<domain_keystore/truststore_file> -storepass
<domain_keystore/truststore _password> -file
<file>.cert -alias <alias_name>
```

Note the SSL Specific Information for the Installer

Make a note of the values for the following SSL prompts that you will have to respond to when you run the Secure@Source installer:

- Select the cluster type: Select External Cluster
- Is Cluster Kerberos-enabled? No
- Select the Hadoop Distribution Type: ClouderaManager, Hortonworks or Azure HDInsight
- Hadoop Cluster URL : Cluster URL
- Hadoop Cluster User Name :
- Hadoop Cluster Password :
- Catalog Service Name: (default is CS) :
- Catalog Service Key Store Location:
- Catalog Service Key Store Password:
- Catalog Service Key Store Alias:
- Is External Cluster SSL Enabled? Yes
- Solr Keystore File:
- Solr Keystore Password:
- Service Cluster Name:
- Catalog Service HTTPS Port:

Prerequisites for Kerberos-Enabled External Cluster

Perform the following steps before you install Secure@Source on a Kerberos-enabled external cluster:

❑ Configure the following properties for the Catalog Service that connects to a Kerberos-enabled external cluster:

- A keytab file that contains all the users in LDAP.
- Kerberos domain name.
- HDFS namenode and YARN Resource Manager service principals

- ❑ Verify that you install the following prerequisite packages before you enable Kerberos:
  - krb5-workstation
  - krb5-libs
  - krb5-auth-dialog
- ❑ Configure the Key Distribution Center (KDC) hostname and IP address on all cluster nodes and domain machines in the `/etc/hosts` file.
- ❑ Ensure that the `krb5.conf` file is located in all cluster nodes and domain machines under the `/etc` directory.
- ❑ Create a local user with username as `<username>` on all the hosts in the cluster. This step is required to launch the application on YARN as the user configured for Catalog Service.
- ❑ The principal user name and service cluster name in HDFS must be the same.
- ❑ Create the following users in the LDAP security domain where `<username>` is the service cluster name.
  - `<username>@KERBEROSDOMAIN.COM`
  - `<username>/<Hadoop hostname>@KERBEROSDOMAIN.COM`

    Note: Create the user ID for all the Hadoop hosts in the cluster.
  - `HTTP/<Hadoop hostname>@KERBEROSDOMAIN.COM`

    Note: Create the user ID for all the Hadoop hosts in the cluster.
  - Create a keytab file with credentials for all these users created in LDAP. You can create keytab files for each one of the users in KDC server and merge them using the ktutil command to create single keytab file.
- ❑ Create `<SecureAtSource_ServiceName>_elasticsearch` folder in `/var/lib/hadoop-yarn` directory. The group must be the same as the principal user and the owner must be the principle user.

  Note: The `<SecureAtSource_ServiceName>_elasticsearch` folder name must be in lower case.

  Example:

  ```
  sudo mkdir /var/lib/hadoop_yarn/sats_elasticsearch
  sudo chown infa:infa /var/lib/hadoop_yarn/sats_elasticsearch
  ```
- ❑ Set the `udp_preference_limit` parameter in the `/etc/krb5.conf` Kerberos configuration file to `1` to always use TCP. This parameter determines the protocol that Kerberos uses when it sends a message to the KDC. The Informatica domain only supports the TCP protocol. If the `udp_preference_limit` parameter is set to any other value, the Informatica domain might shut down unexpectedly.

  Set the `udp_preference_limit` on both the Domain and Hadoop machines.

Note: Secure@Source does not support deployment on a Hortonworks version 2.6 cluster where Kerberos is enabled. However, if you have Hortonworks 2.6, you can use a workaround. See the H2L article *Installing Secure@Source on a Kerberized Hortonworks 2.6 External Cluster* located on the Informatica Network at https://search.informatica.com.

Note the Kerberos Information for the Installer

Make a note of the values for the following Kerberos prompts you will have to respond to when you run the Secure@Source installer:

- Is Cluster Kerberos-enabled? **Yes**
- Select the Hadoop Distribution Type and provide the cluster details.
- Service Cluster Name: Must be the same as the principal user.
- KDC Domain Name: Specify the KDC Domain name for example: `HADOOP.COM`
- Keytab Location: Specify the keytab file location with filename.

- Kerberos Configuration File location: Specify the location with file name of the Kerberos configuration file. Example: `etc/krb5.conf`

Deploying Secure@Source on an Existing Hortonworks or Cloudera Cluster that is not SSL Enabled

To deploy Secure@Source on an existing Hortonworks or Cloudera cluster that is not SSL enabled, perform the following steps:

1. Configure the following properties in the `/etc/hadoop/conf/ssl-client.xml` file:
   - `ssl.client.truststore.location`
   - `ssl.client.truststore.password`
2. Place the domain truststore file in all the Hadoop nodes in the `/opt` directory. For example: `/opt/truststore/infa_truststore.jks`
3. Ensure that the `ssl.client.truststore.location` value is set to the `/opt` directory and not the `/etc` directory. Verify that you configure the full path to the truststore file for the `ssl.client.truststore.location` property. For example, you can set the value similar to `/opt/truststore/infa_truststore.jks`.

Step 6. Secure@Source and Hadoop Ports

For established Informatica port numbers, see the *Configuring Ports for Big Data Management, Data Integration Hub, Enterprise Information Catalog, and Intelligent Data Lake* H2L located on the Informatica Knowledge base at https://kb.informatica.com.

Verify that the following TCP and UDP ports are open:

Interface Source	Interface Target	TCP Ports	UDP Ports
Laptop/Workstation, Hadoop Host	App Host {Secure at Source}	6005 through 6009, 6014-6114, 6643, 8443, 9443, 9553, 18295	
Hadoop Host	App Host {Secure at Source}	6013, 6113, 9085, 9090,15380, 15381, 15385, 15386, 18105, 18095, 52652	
App Host {Secure at Source}, Laptop/ Workstation	Hadoop Host	8042, 8044, 8080, 8088, 8090, 19888	
App Host {Secure at Source}	Hadoop Host	2181, 5432, 6188, 6443, 6605, 8010, 8020, 8025, 8030, 8032, 8141, 8188, 8190, 8245, 8440, 8441, 8480, 8481, 8670, 9553, 10020, 19888, 45454, 50010, 50070, 50075, 50090, 50470, 50475, 59925, 60200	59804
App Host {Secure at Source}, Hadoop Host	Repository database server	For example: 1521 for Oracle, 1433 for Microsoft SQL Server	

Step 7. Verify Prerequisites for the Repository Database

Create Users

Create users for the following components:

Component	Tablespace Name
Domain Repository	INFA_DOMAIN
Model Repository	INFA_MRS
Model Repository Monitoring Statistics Service	INFA_MRSMS
Content Management	INFA_CMS
Profiling Warehouse	INFA_DIS
Secure@Source	INFA_SATS
Persistent Data Masking*	INFA_PDM
*Required if you plan to enable Persistent Data Masking.	

IBM Db2 Repository Database Requirements

Setup the national character mapping configuration parameter, NCHAR_MAPPING, to GRAPHIC_CU16.

You can use the following commands:

```
db2 connect to <Secure@Source database name>
db2 update db cfg using NCHAR_MAPPING GRAPHIC_CU16
db2 connect reset
```

Microsoft SQL Server Repository Database Requirements

Complete the following tasks if the Secure@Source repository database is Microsoft SQL Server:

Database Settings

Important: The database collation type must be case sensitive. For example: `COLLATE SQL_Latin1_General_CP1_CS_AS`

Configure the following settings:

- ❑ READ_COMMITTED_SNAPSHOT: ON
  This minimizes locking contention.
- ❑ ALLOW_SNAPSHOT_ISOLATION: ON
- ❑ AUTO_UPDATE_STATISTICS_ASYNC: ON
- ❑ AUTO_UPDATE_STATISTICS: ON

Tablespace Sizes

Assign the following tablespace sizes:

Component	Tablespace Name	Size
Domain Repository	INFA_DOMAIN	8 GB
Model Repository	INFA_MRS	75 GB
Model Repository Monitoring Statistics Service	INFA_MRSMS	25 GB
Content Management	INFA_CMS	25 GB
Data Integration Service and Profiling Warehouse	INFA_DIS	250 GB
Secure@Source	INFA_SATS	50 GB
Persistent Data Masking*	INFA_PDM	25 GB
*Required if you plan to enable Persistent Data Masking.		

User Role

Each user must have the server role `public`.

User Privileges

Each user must have the following privileges:

- ❑ ALTER
- ❑ CONNECT
- ❑ CREATE FUNCTION
- ❑ CREATE SCHEMA
- ❑ CREATE TABLE
- ❑ CREATE VIEW
- ❑ DELETE
- ❑ EXECUTE
- ❑ INSERT
- ❑ REFERENCES
- ❑ SELECT
- ❑ UPDATE

Oracle Repository Database Requirements

Complete the following tasks if the Secure@Source repository database is Oracle:

Database Settings

Configure the following settings:

- ❑ open_cursors parameter: 4000
- ❑ Sessions and processes parameters: 1500
- ❑ SGA: 4000 MB or higher

Tablespace Sizes

Assign the following tablespace sizes:

Component	Tablespace Name	Size
Domain Repository	INFA_DOMAIN	8 GB
Model Repository	INFA_MRS	75 GB
Model Repository Monitoring Statistics Service	INFA_MRSMS	25 GB
Content Management	INFA_CMS	25 GB
Data Integration Service and Profiling Warehouse	INFA_DIS	250 GB
Secure@Source	INFA_SATS	50 GB
Persistent Data Masking*	INFA_PDM	25 GB
*Required if you plan to enable Persistent Data Masking.		

User Privileges

Each user must have the following privileges:

- ❑ CREATE PROCEDURE
- ❑ CREATE SESSION
- ❑ CREATE SEQUENCE
- ❑ CREATE SYNONYM
- ❑ CREATE TABLE
- ❑ CREATE TRIGGER
- ❑ CREATE VIEW

User Resource Limits

Assign the following resource limits:

- ❑ SESSIONS_PER_USER: 100 or higher
- ❑ CPU_PER_SESSION: unlimited
- ❑ CONNECT_TIME: unlimited
- ❑ IDLE_TIME: unlimited

Step 8. Install Secure@Source

Log in as the Informatica user and perform the following tasks to install Secure@Source:

- ❑ Download the Secure@Source software and copy it to the Secure@Source and Data Integration Service servers.
- ❑ Download the Informatica license file and copy it to the same location as the installation software.
- ❑ Download the `ScannerBinaries.zip` file to the `<Secure@Source_Installer_Location>/Server/LDM/source` directory.

Note: Ensure that you download the binaries file for the correct operating system and version.

❑ Run the Secure@Source installer.

For details on running the Secure@Source installer, see the *Informatica Secure@Source Installation Guide*.

Step 9. Configure Connectivity to the Data Source

Connect to the data source that you want to scan. You might need to install or configure additional components to connect to the data source. Then enable the ports to the data source on the Secure@Source and all Hadoop nodes. For example, enable port 1521 for Oracle sources and port 1433 for Microsoft SQL Server sources.

Category	Data Store	Connectivity Requirements
Application Sources	SAP	- Import SAP transports. - Install SAP JCo.
Big Data Sources	Cloudera Navigator	None.
	Hadoop Distributed File System	Create a cluster configuration object in the Administrator Tool.
	Hive	Create a cluster configuration object in the Administrator Tool.
Cloud Sources	Amazon Redshift	- Download the Amazon Redshift JDBC Driver. - Obtain the JDBC URL. - Configure a JDBC Connection With Apache Maven.
	Amazon S3	- Enable account with Amazon Web Services. - Note the Access Key and Secret Key.
	Microsoft Azure SQL Database	Note the Azure Blob Storage name and key.
	Microsoft Azure SQL Data Warehouse	Copy the jar file `mssql-jdbc-6.2.0.jre8.jar` to the following directory: `INFA_HOME/java/lib/ext`
	Salesforce	- Verify Salesforce user account and privileges. - Register the Salesforce SSL certificates with the Informatica domain.
Data Integration	Informatica Big Data Management	- Copy the Informatica Big Data Management jar files to the following directory: `<INFA_HOME>/services/SecureAtSourceService/ROOT/WEB-INF/eclipse/plugins` - Create a Secure@Source temporary folder in the following directory: `<INFA_HOME>/tomcat/temp/` - Restart the Secure@Source Services.
	Informatica Cloud	Import the certificate for Informatica Cloud into the Secure@Source trust store.
	Informatica PowerCenter on IBM DB2 Database	Get the sitekey file.

Category	Data Store	Connectivity Requirements
	Informatica PowerCenter on Microsoft SQL Server Database	Get the sitekey file.
	Informatica PowerCenter on Oracle Database	Get the sitekey file.
	Microsoft SQL Server Integration Services	For repository based service: - Configure the 10.2 supported MITI Agent URL to run on the Microsoft SQL Server Integration Services server. - Assign administrator privileges to the MITI Agent. For file based service: - Configure the 10.2 supported MITI Agent URL to run on any server, client, or accessed box. - Copy the Microsoft SQL Server Integration Services mapping files from Visual Studio to any directory on the Secure@Source domain box.
Database Management Sources	IBM DB2	- Install the database client software. - Configure native connectivity to the database.
	IBM DB2 i5	Install the PowerExchange client.
	IBM DB2 zOS	Install the PowerExchange client.
	Microsoft SQL Server	- Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
	MySQL	- Install the database client software or install the JDBC and ODBC drivers. - Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
	Netezza	- Install the database client software or install the JDBC and ODBC drivers. - Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
	Oracle	- Install the database client software. - Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
	Sybase	- Install the database client software or install the JDBC driver. - Configure ODBC or JDBC connectivity to the database. - Configure permissions and privileges for the database user account.
	Teradata	- Install the database client software or install the JDBC and ODBC drivers. - Configure either native mode or Hadoop mode connectivity to the database. - Configure permissions and privileges for the database user account.
File Management Sources	File System	Create a mount point to the file system on the Hadoop and Informatica domain nodes.

For details on how to connect to each data source, see the *Informatica Secure@Source Administrator Guide*.

Author